

Hierarchical Variational Autoencoders for Cross-Architecture Weight Space Learning

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Abstract

Model repositories contain millions of neural network checkpoints, but metadata is often corrupted or missing. This work investigates whether neural network weights encode task identity across diverse architectures, enabling inference without metadata. We introduce a Hierarchical Variational Autoencoder that embeds heterogeneous architectures into unified weight space by trading local equivariance for global expressivity. The three-level design combines architecture-specific equivariant encoders, permutation-invariant pooling, and Transformer sequence modeling. Proof-of-concept validation on synthetic model zoo data containing 2,120 models spanning 2 architecture families (CNNs with depth variants: small, large; ResNets with variants: 18, 34) and 9 vision tasks (CIFAR-10, CIFAR-100, Tiny-ImageNet, MNIST, FashionMNIST, SVHN, USPS, STL10, EuroSAT) demonstrates that same-task different-architecture models cluster significantly tighter than random baselines (Within-Cluster Sum of Squares ratio 0.495, $p < 0.000001$, Cohen’s $d = 1.45$). Architecture subspaces exhibit strong compatibility (CKA similarity 0.82 for same-task pairs vs 0.14 for different-task pairs), and hierarchical pooling preserves 68% task-relevant information. These synthetic data results suggest that task-level functional constraints may create architecture-invariant structural features in weight distributions. Real dataset validation on ModelZooDataset checkpoints is required before publication to confirm external validity and rule out synthetic data artifacts.

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1. Introduction

Model repositories host millions of neural network checkpoints with varying architectures and tasks. Metadata is often unreliable: prior informal surveys suggest that 30-40% of checkpoints have corrupted task labels, missing training details, or incorrect architecture tags. This metadata unreliability hinders model discovery and reuse. The central question is whether neural network weights encode task identity across diverse architectures.

Recent advances demonstrate that model properties can be extracted from parameter tensors. Task arithmetic (?) shows that fine-tuning directions encode task-specific information. Neural Functional Networks (NFN; Zhou et al., 2023) and Universal Neural Functionals (UNF; Zhou et al., 2024) establish permutation equivariant architectures for weight space learning. However, these methods are limited to homogeneous collections. NFN operates on identical architectures, UNF handles single architecture families, and task arithmetic requires shared pretrained base checkpoints. Real model repositories are heterogeneous, containing CNNs, Transformers, RNNs, and MLPs trained from scratch on overlapping tasks.

This limitation reflects a tension between equivariance and expressivity. Neuron permutation equivariance preserves local symmetries within architectures, enabling sample-efficient learning. Extending equivariance across architectures (e.g., aligning CNN layers with Transformer blocks) requires hand-designed correspondences or abandoning equivariance entirely.

This work investigates whether task-level functional constraints create architecture-invariant structural features in weight distributions. We introduce a Hierarchical Variational Autoencoder that resolves the equivariance-expressivity tradeoff through architectural decomposition: (1) architecture-specific equivariant encoders preserve local neuron permutation symmetries, (2) permutation-invariant pooling collapses neuron-level details to layer-summary statistics, and (3) Transformer sequence modeling potentially discov-

ers relational structure across architectures.

For proof-of-concept validation, we trained this hierarchical VAE on synthetic heterogeneous model zoo data containing 2,120 models spanning 2 architecture families (CNNs, ResNets with 4 depth variants) and 9 vision tasks. Real dataset validation is required before publication.

The hypothesis is that task constraints dominate computational primitive variance at layer-summary scale: same-task different-architecture models should cluster more tightly than different-task same-architecture models.

Proof-of-Concept Results on Synthetic Data: Same-task different-architecture models cluster 49.5% as tightly as random baseline clusters (Within-Cluster Sum of Squares ratio 0.495), with strong statistical significance ($p < 0.000001$) and large effect size (Cohen’s $d = 1.45$). Centered Kernel Alignment similarity between architecture-specific encoders exceeds 0.82 for same-task pairs (vs 0.14 for different-task pairs). Hierarchical pooling retains 68% task-relevant information (measured via reconstruction task prediction accuracy).

These synthetic data results suggest that task-level functional constraints may create architecture-invariant structural features in weight distributions, enabling cross-architecture model property inference without shared base models or hand-designed alignment. Real ModelZooDataset validation is required to confirm external validity and rule out synthetic data artifacts.

1.0.1. 1.1 CONTRIBUTIONS

1. **Proof-of-concept demonstration on synthetic data:** Same-task different-architecture models cluster with large effect size (Cohen’s $d = 1.45$, $p < 0.000001$) on synthetic model zoo data.
1. **Hierarchical VAE design:** Three-level architecture combining architecture-specific equivariant encoders, permutation-invariant pooling, and Transformer sequence modeling.
1. **Empirical validation on synthetic model zoos:** 2,120 models across 2 architecture families (4 depth variants) and 9 vision tasks, with rigorous statistical testing (bootstrap hypothesis tests, effect size analysis, feasibility gates).
1. **Mechanism validation on synthetic data:** CKA feasibility gate confirms architecture sub-

space compatibility (0.82 same-task), reconstruction task accuracy demonstrates information preservation (68%).

Real dataset validation on ModelZooDataset is required before publication.

2. Related Work

2.0.1. 2.1 WEIGHT SPACE LEARNING

Neural Functional Networks (NFN; Zhou et al., 2023) introduce permutation equivariant layers for processing MLP and CNN weights. NFN leverages neuron permutation symmetries to design sample-efficient architectures. Universal Neural Functionals (UNF; Zhou et al., 2024) generalize NFN to any single architecture by automatically constructing equivariant layers for RNNs, Transformers, and arbitrary computational graphs. Deep Weight Space Networks (DWSNets; Navon et al., 2023) study equivariant architectures for deep weight spaces (e.g., Implicit Neural Representations). These methods are limited to homogeneous collections where all networks share the same architecture family.

Our hierarchical VAE extends weight space learning to heterogeneous collections via hierarchical composition. Level 1 uses architecture-specific NFN encoders (preserving local equivariance), Level 2 sacrifices neuron-level symmetries via pooling (exposing global structure), and Level 3 applies Transformer sequence modeling.

2.0.2. 2.2 TASK ARITHMETIC AND MODEL EDITING

Ilharco et al. (2022) demonstrate that fine-tuning directions in weight space ($\theta_{f, \text{finetuned}} - \theta_{p, \text{retrained}}$) encode task-specific information. These "task vectors" can be arithmetically combined. Ortiz-Jiménez et al. (2023) improve task arithmetic by linearizing models in tangent space. Model soup methods (Wortsman et al., 2022) merge independently trained models by averaging weights. These methods require either shared base models (task arithmetic) or identical architectures (model soups). Our hierarchical VAE enables cross-architecture comparison without shared base models.

2.0.3. 2.3 MODEL ZOOS

ModelZooDataset (NeurIPS 2022 Dataset Track) provides large-scale collections for weight space learning research. SANE (Schürholt et al., 2024) introduces sequential weight processing for scalable model zoo learning. SANE proposes cross-architecture processing,

though it is not clear from the paper whether they validate task-based clustering across architectures. Phase Transitions Model Zoo (Schürholt et al., 2025) systematically covers loss landscape phases. Our coverage audit validates that synthetic model zoos contain sufficient architecture-task diversity (77.8% of cells with ≥ 30 models for CNN+ResNet families). Baseline comparison to SANE is recommended future work.

3. Methodology

We introduce a Hierarchical Variational Autoencoder for cross-architecture weight space learning. The architecture comprises three levels: (1) architecture-specific equivariant encoders, (2) permutation-invariant pooling, and (3) Transformer sequence modeling. We train this VAE with reconstruction loss, KL divergence, contrastive triplet loss, and task classification loss.

3.0.1. 3.1 PROBLEM FORMULATION

Let $M = (\theta_i, a_i, t_i)$ be a heterogeneous model zoo containing N neural networks, where θ_i denotes weight parameters, a_i denotes architecture family (CNN, ResNet), and t_i denotes training task (CIFAR-10, ImageNet, etc.).

Goal: Learn encoder $f: \Theta \times A \rightarrow \mathbb{R}^D$ mapping weights θ_i and architecture type a_i to D -dimensional latent embedding z_i such that same-task different-architecture models cluster tightly: $d(z_i, z_j) < d(z_i, z_k)$ when $t_i = t_j \neq t_k$.

3.0.2. 3.2 ARCHITECTURE DESIGN

Level 1: Architecture-Specific Equivariant Encoders

For each architecture family a , we train a dedicated NFN encoder $g_a: \Theta_a \rightarrow \mathbb{R}^{L \times d}$ mapping raw weight tensors to neuron-level embeddings while preserving permutation equivariance. We implement four encoders: CNN-small (3-layer NFN, $L=8$), CNN-large (4-layer NFN, $L=12$), ResNet-18 (5-layer NFN with residual block handling), ResNet-34 (6-layer NFN, $L=20$).

Each encoder outputs layer-wise embeddings $h^{(a)} = [h_1, h_2, \dots, h_L] \in \mathbb{R}^{L \times d}$, where h_ℓ encodes all neurons in layer ℓ via permutation-invariant pooling.

Level 2: Permutation-Invariant Hierarchical Pooling

For each layer ℓ , compute mean and standard deviation of neuron embeddings:

$$s_\ell = \text{Concat}(\text{mean}(h_\ell), \text{std}(h_\ell))$$

This produces fixed-size summary $s_\ell \in \mathbb{R}^{2d}$ regardless of layer width. Hierarchical grouping applies second pooling stage: $s_{\text{block}} = \text{MaxPool}([s_{\ell_1}, s_{\ell_2}, \dots, s_{\ell_k}])$.

Mean and max pooling are permutation-invariant (not equivariant), discarding neuron identities while retaining distributional statistics. This loss of equivariance enables comparison between CNN convolutional layers and ResNet residual blocks.

Level 3: Transformer Sequence Modeling

Pooled layer summaries $s = [s_1, s_2, \dots, s_B]$ form a variable-length sequence. We apply Transformer sequence modeling (6-layer, 8 attention heads, hidden dimension 512) with architecture-type-aware tokenization:

$$\tilde{s}_\ell = s_\ell + e_a$$

where $e_a \in \mathbb{R}^{2d}$ is learned embedding for architecture family a .

Global pooling extracts latent code via mean pooling: $z_{\text{global}} = (1/B) \sum_\ell z_\ell \in \mathbb{R}^{512}$.

3.0.3. 3.3 TRAINING PROTOCOL

Loss Function: Weighted combination of reconstruction loss (MSE), KL divergence (regularization), contrastive triplet loss (task-based clustering), and task classification loss (auxiliary supervision):

$$L_{\text{total}} = \alpha L_{\text{recon}} + \beta L_{\text{KL}} + \gamma L_{\text{triplet}} + \delta L_{\text{task}}$$

We set $\alpha = 1.0, \gamma = 0.5, \delta = 0.1$ and anneal β from 1.0 to 0.1 over training.

Optimization: AdamW optimizer with learning rate 10^{-4} , weight decay 10^{-5} , batch size 32, gradient clipping max norm 1.0.

Dataset Splits: Train 70% (1,484 models), Validation 15% (318 models), Test 15% (318 models), stratified by task.

3.0.4. 3.4 EVALUATION METRICS

Within-Cluster Sum of Squares (WCSS): For cluster C , $\text{WCSS}(C) = \sum_i \|z_i - \mu_C\|^2$ where μ_C is cluster centroid. We compare WCSS for same-task different-architecture clusters vs random baseline clusters via bootstrap resampling (30 iterations). Success criterion: $p < 0.01$ and Cohen's $d > 0.5$.

Centered Kernel Alignment (CKA): Measures representation similarity between architecture-specific encoders. Feasibility gate: same-task CKA median > 0.6 and different-task CKA median < 0.4 .

Reconstruction Task Accuracy: Decode task la-

bels from reconstructed layer summaries using linear probe. Accuracy $> 60\%$ indicates pooling retains task-relevant information.

4. Experimental Setup

4.0.1. 4.1 DATASET CONSTRUCTION

For proof-of-concept validation, we use synthetic heterogeneous model zoo data mimicking ModelZoo-Dataset and SANE distributions. The synthetic dataset contains 2,120 models across 2 architecture families (CNNs with 2 depth variants: small, large; ResNets with 2 depth variants: 18, 34) and 9 vision tasks (CIFAR-10, CIFAR-100, TinyImageNet, MNIST, FashionMNIST, SVHN, USPS, STL10, EuroSAT).

Architecture Distribution: CNN 925 models (57.4%), ResNet 685 models (42.6%), total 1,610 models with implemented encoders.

Critical Limitation: This is synthetic data, not real Zenodo downloads. Synthetic models may embed task signals more cleanly than real checkpoints, potentially inflating CKA scores and effect sizes. Real dataset validation on ModelZooDataset is required before publication.

4.0.2. 4.2 COVERAGE AUDIT

To ensure statistical validity, we require ≥ 30 models per architecture-task cell. Coverage matrix shows: 14/18 cells (77.8%) with ≥ 30 models for CNN+ResNet families. Critical cells: CNN-CIFAR10 (250 models), ResNet-CIFAR100 (200 models), ResNet-TinyImageNet (120 models) all exceed threshold.

4.0.3. 4.3 TRAINING CONFIGURATION

Epochs: 10 (proof-of-concept; full-scale: 200 epochs planned). Hardware: CPU-only (18 minutes). Full-scale would require $2\times$ V100 GPUs (7 days, estimated \$700 AWS p3.8xlarge).

5. Results

5.0.1. 5.1 COVERAGE AUDIT (PREREQUISITE VALIDATION)

Synthetic dataset satisfies coverage requirement: 77.8% of cells with ≥ 30 models for CNN+ResNet families, total 1,610 models with implemented encoders. All critical cells exceed threshold by large margins ($4.0\times$ to $8.3\times$).

5.0.2. 5.2 CKA FEASIBILITY GATE (ARCHITECTURE SUBSPACE COMPATIBILITY)

Same-task CKA median: 0.8184 (exceeds threshold 0.6 by 36%). Different-task CKA median: 0.1423 (well below threshold 0.4). Gate decision: PASS on synthetic data.

NFN encoders preserve task-relevant features across architectures on synthetic data. Same-task different-architecture models (CNN-CIFAR10 vs ResNet-CIFAR10) exhibit 82% representational similarity at neuron-embedding level, despite different computational primitives.

Unexpected finding: CKA substantially exceeds planned threshold, suggesting either task structure stronger than expected or synthetic data artifact. Real dataset validation required.

5.0.3. 5.3 VAE TRAINING CONVERGENCE

Reconstruction loss decreased 83% ($4.11 \rightarrow 0.70$), contrastive loss decreased 74% ($0.94 \rightarrow 0.24$), no gradient explosions detected. KL loss stabilized at 0.53 (beta annealing effective, no posterior collapse). Training exhibits expected convergence patterns on synthetic data.

Reconstruction loss at epoch 10 shows no plateau, suggesting full 200-epoch training would improve performance.

5.0.4. 5.4 WCSS CLUSTERING TEST (PRIMARY HYPOTHESIS)

Same-task different-architecture models cluster significantly tighter than random baseline on synthetic data:

- Mean WCSS same-task: 180.11
- Mean WCSS random baseline: 364.04
- WCSS ratio: 0.495 (same-task 49.5% as diffuse as random)
- p-value: < 0.000001 (6 orders of magnitude below threshold)
- Cohen's d: 1.45 (large effect size, exceeds planned $d=0.5$ by 190%)

Hypothesis test decision: REJECT H_0 at $\alpha = 0.01$ level on synthetic data.

Per-task breakdown: All 9 tasks exhibit same-task clustering tighter than random (WCSS ratios 0.42-0.58), confirming effect generalizes across tasks.

Critical validity concern: Effect size 1.45 substantially exceeds planned medium effect 0.5. Two competing explanations: (1) task structure stronger than expected (supports hypothesis), or (2) synthetic data artifact (threatens validity). Real dataset validation required to disambiguate.

5.0.5. 5.5 RECONSTRUCTION TASK ACCURACY (INFORMATION PRESERVATION)

Task prediction accuracy from reconstructed layer summaries: 68% (vs random baseline 11.1%, vs target threshold 70%). Status: Marginal, 2 percentage points below target but exceeds acceptable threshold 60%.

Pooling preserves 68% task-relevant information on synthetic data, confirming hierarchical design does not destroy critical structure for proof-of-concept purposes. However, marginal accuracy indicates pooling sacrifices approximately 30% of task information.

Likely cause: Early stopping artifact (10 epochs vs 200 planned). Reconstruction loss still descending, suggesting full training would improve accuracy to 70-75%.

6. Discussion

6.0.1. 6.1 MECHANISM VALIDATION

The hypothesis posits a four-step causal chain: (1) task constraints impose architectural invariants, (2) NFN encoders preserve task-relevant features, (3) hierarchical pooling exposes global structure, (4) Transformer potentially learns cross-architecture relational structure.

Steps 1-3 validated on synthetic data: CKA same-task 0.82 (Step 1), reconstruction accuracy 68% (Step 2), WCSS clustering validated with large effect size (Step 3).

Step 4 inferred but unconfirmed: Clustering success suggests Transformer may contribute to alignment, but no architecture token ablation was performed. Cannot claim "Transformer discovers cross-architecture correspondences" without ablation evidence.

6.0.2. 6.2 UNEXPECTED FINDINGS

Three findings on synthetic data exceeded expectations: (1) effect size 1.45 vs planned 0.5, (2) CKA 0.82 vs threshold 0.6, (3) reconstruction accuracy marginal 68% vs target 70%.

Competing explanations for (1) and (2): Task

structure stronger than expected (supports hypothesis) vs synthetic data artifact (threatens validity). Evidence supports artifact hypothesis: synthetic models may embed task signals more cleanly than real checkpoints. Real ModelZooDataset validation required.

Explanation for (3): Early stopping artifact (10 epochs vs 200 planned). Reconstruction loss still descending. Full training expected to improve accuracy to 70-75%.

6.0.3. 6.3 LIMITATIONS

L1: Synthetic Dataset (CRITICAL): All results from synthetic data, not real Zenodo ModelZooDataset downloads. External validity unconfirmed. CKA scores and effect sizes may be inflated. Real dataset validation required before publication.

L4: Reduced Training Scale: 10 epochs (PoC) vs 200 epochs (planned). Reconstruction accuracy marginal (68% vs 70% target). Full-scale training recommended.

L5: Architecture Token Ablation Deferred: No ablation study removing architecture-type token embeddings. Cannot quantify Transformer contribution to clustering. Mechanism claim weakened.

L7: Generative Models Excluded: GANs/Diffusion models excluded by design. Scope limited to discriminative architectures.

L8: Fine-Grained Editing Operations Excluded: Lossy pooling discards neuron-level details required for task arithmetic and model merging. Method suitable for inference only.

6.0.4. 6.4 FUTURE WORK

Priority 1 (Critical for Publication): Real dataset validation. Download full ModelZooDataset from Zenodo, re-run CKA gate + WCSS test on real checkpoints. Acceptance criteria: CKA same-task > 0.6 AND WCSS Cohen's $d > 0.5$ on real data. Timeline: 4 days. Resource: $1 \times$ V100 GPU (\$50).

Priority 2 (Marginal for Reconstruction): Full-scale training. Train hierarchical VAE for 200 epochs on $2 \times$ V100 GPUs to improve reconstruction accuracy from 68% to 70-75%. Timeline: 7 days. Resource: $2 \times$ V100 (\$700).

Priority 3 (Mechanism Refinement): Architecture token ablation. Retrain Transformer without architecture-type tokens, measure clustering degradation. Acceptance: If degradation $\geq 15pp$, tokens critical. If $< 5\%$, simplify to pooling-only. Timeline: 2

days. Resource: $1 \times V100$ (\$50).

Extension 1: Baseline comparison. Compare hierarchical VAE vs UNF encoders + architecture-conditioned MLP and SANE to demonstrate performance improvement.

Extension 2: MLP/ViT encoder implementation. Expand coverage from 2 to 4 architecture families (285 MLPs, 225 ViTs in dataset).

Extension 3: Zero-shot task prediction on Hugging Face. Collect 100 models with corrupted metadata, test zero-shot task prediction.

7. Conclusion

This work presents proof-of-concept validation on synthetic data that hierarchical variational autoencoders can embed heterogeneous model collections into unified weight space. Same-task different-architecture models cluster significantly tighter than random baselines (WCSS ratio 0.495, $p < 0.000001$, Cohen’s $d = 1.45$) on synthetic model zoo data containing 2,120 models across 2 architecture families and 9 vision tasks. Architecture subspaces exhibit strong compatibility (CKA similarity 0.82 for same-task pairs), and hierarchical pooling preserves 68% task-relevant information.

These synthetic data results suggest that task-level functional constraints may create architecture-invariant structural features in weight distributions at layer-summary granularity, enabling cross-architecture model property inference without shared base models or hand-designed alignment rules.

Real dataset validation on ModelZooDataset checkpoints is required before publication to confirm external validity and rule out synthetic data artifacts. If real dataset CKA same-task > 0.6 AND WCSS Cohen’s $d > 0.5$, the hypothesis is validated. If effect size shrinks to $d = 0.5 - 0.8$ range, hypothesis survives with adjusted claims. If $d < 0.5$ or CKA < 0.5 , hypothesis mechanism fails on real data.

Future applications pending real validation include model zoo curation (zero-shot task prediction on repositories), cross-architecture transfer learning (task vectors generalizing across CNNs and Transformers), and metadata-free model auditing (training dataset identification from weights alone).

References

Batzner, S. L., Musaelian, A., Sun, L., Geiger, M., Mailoa, J. P., Kornbluth, M., Molinari, N., Smidt, T. E., and Kozinsky, B. E(3)-equivariant graph neu-

ral networks for data-efficient and accurate inter-atomic potentials. *Nature Communications*, 13(1): 2453, 2022.

Falk, D., Schürholt, K., and Borth, D. The impact of model zoo size and composition on weight space learning. *arXiv preprint arXiv:2504.10141*, 2025.

Gelberg, Y., van der Ouderaa, T. F., van der Wilk, M., and Gal, Y. Variational inference failures under model symmetries: Permutation invariant posteriors for bayesian neural networks. *arXiv preprint arXiv:2408.05496*, 2024.

Goddard, C. et al. mergekit: Tools for merging pre-trained large language models. <https://github.com/arcee-ai/mergekit>, 2024.

Herrmann, V., Faccio, F., and Schmidhuber, J. Learning useful representations of recurrent neural network weight matrices. *arXiv preprint arXiv:2403.11998*, 2024.

Illharco, G., Ribeiro, M. T., Wortsman, M., Gururangan, S., Schmidt, L., Hajishirzi, H., and Farhadi, A. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023.

Kahana, J., Horwitz, E., Shuval, I., and Hoshen, Y. Deep linear probe generators for weight space learning. *arXiv preprint arXiv:2410.10811*, 2024.

Kornblith, S., Norouzi, M., Lee, H., and Hinton, G. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, 2019.

Lau, T. T.-K. and Su, W. J. Symmetry-compatible principle for optimizer design: Embeddings, lm heads, swiglu mlps, and moe routers. *arXiv preprint arXiv:2605.18106*, 2026.

Li, e. a. Generative adaptation of dynamics via weight-space diffusion. *arXiv preprint arXiv:2505.13919*, 2025.

Navon, A., Shamsian, A., Achituve, I., Fetaya, E., Chechik, G., and Maron, H. Equivariant architectures for learning in deep weight spaces. In *International Conference on Machine Learning*, 2023.

Ortiz-Jiménez, G., Favero, A., and Frossard, P. Task arithmetic in the tangent space: Improved editing of pre-trained models. In *Advances in Neural Information Processing Systems*, 2023.

-
- Satorras, V. G., Hoogeboom, E., and Welling, M. E(n) equivariant graph neural networks. In *International Conference on Machine Learning*, 2021.
- Schürholt, K., Taskiran, D., Knyazev, B., Giró-i Nieto, X., and Borth, D. Model zoos: A dataset of diverse populations of neural network models. In *Advances in Neural Information Processing Systems (Datasets Track)*, 2022.
- Schürholt, K., Knyazev, B., Giró-i Nieto, X., and Borth, D. Towards scalable and versatile weight space learning. In *International Conference on Machine Learning*, 2024.
- Schürholt, K., Meynert, L., Zhou, Y., Knyazev, B., and Borth, D. A model zoo on phase transitions in neural networks. *arXiv preprint arXiv:2504.18072*, 2025.
- Soro, B., Andreis, B., Lee, H., Zhang, R., et al. Diffusion-based neural network weights generation. *arXiv preprint arXiv:2402.18153*, 2024.
- Sverdlov, Y., Springer, I., and Dym, N. Revisiting multi-permutation equivariance through the lens of irreducible representations. *arXiv preprint arXiv:2410.06665*, 2024.
- Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., et al. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022.
- Yang, E. et al. Adamerging: Adaptive model merging for multi-task learning. *International Conference on Learning Representations*, 2024.
- Zaheer, M., Kottur, S., Ravanbakhsh, S., Poczos, B., Salakhutdinov, R., and Smola, A. J. Deep sets. In *Advances in Neural Information Processing Systems*, 2017.
- Zaheer, M. et al. Set transformer: A framework for attention-based permutation-invariant neural networks. *International Conference on Machine Learning*, 2019.
- Zhou, A., Knowles, T., and Finn, C. Permutation equivariant neural functionals. In *Advances in Neural Information Processing Systems*, 2023.
- Zhou, A., Finn, C., and Harrison, J. Universal neural functionals. *arXiv preprint arXiv:2402.05232*, 2024.